1. Attempts on Relativistic QM

In the case of **special relativity**, physics looks the same on every inertial frames. The spacetime coordinate (contravariant) is introduced as:

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \boldsymbol{x}).$$

Minkowski metric, also known as flat spacetime tensor metric, is defined as:

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

So, the covariant form of the spacetime coordinate is just $x_\mu = g_{\mu\nu}x^\nu$. Another important property of the tensor metric is:

$$g^{\mu\nu}g_{\nu\rho} = \delta^\mu_\nu$$

which is just the delta Kronecker function.

Lorentz transformation is defined by these sets of equations.

$$\begin{aligned} c\bar{t} &= \gamma ct - \gamma\beta x, \\ \bar{x} &= -\gamma\beta ct + \gamma x, \\ \bar{y} &= y, \\ \bar{z} &= z. \end{aligned}$$

Then we can define Lorentz matrix as:

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Now, one can relate a transformation for the spacetime coordinate can be written as:

$$\bar{x}^\mu = \Lambda^\mu_\nu x^\nu + a^\mu,$$

where a^μ is just a constant translation vector. Lorentz transformation matrix must obey this equation:

$$g_{\mu\nu}\Lambda^\mu_\rho\Lambda^\nu_\sigma = g_{\rho\sigma},$$

which is straightforward to prove by contracting the indices μ and ν, we are then left with $\left(-\Lambda^0_\rho\Lambda^0_\sigma + \Lambda^1_\rho\Lambda^1_\sigma + \Lambda^2_\rho\Lambda^2_\sigma + \Lambda^3_\rho\Lambda^3_\sigma\right)$ which we can set as an unknown 4×4 matrix. If we solve for each matrix element, we will get that the matrix is equal to the tensor metric.

The equation ensures that the "length" between two points are the same on either frames.

$$
\begin{aligned}
(\bar{x}' - \bar{x})^2 &= g_{\mu\nu}(\bar{x}' - \bar{x})^\mu (\bar{x}' - \bar{x})^\nu \\
&= g_{\mu\nu}\Lambda^\mu_\rho \Lambda^\nu_\sigma (x' - x)^\rho (x' - x)^\sigma \\
&= g_{\rho\sigma}(x' - x)^\rho (x' - x)^\sigma \\
&= (x' - x)^2.
\end{aligned}
$$

In quantum mechanics, the time evolution of the state of a system is governed by one equation, the Schrodinger equation.

$$
i\hbar \frac{\partial}{\partial t}|\psi,\, t\rangle = H|\psi,\, t\rangle.
$$

Obvious attempt to consider the relativistic version is by setting the hamiltonian using the energy-momentum relation.

$$
\begin{aligned}
E^2 &= (pc)^2 + \left(mc^2\right)^2, \\
H &= +\sqrt{p^2c^2 + m^2c^4}.
\end{aligned}
$$

We can expand the hamiltonian using binomial expansion.

$$
\begin{aligned}
H &= +mc^2\sqrt{1 + \frac{p^2}{m^2c^2}} \\
&= +mc^2\left(1 + \frac{1}{2}\frac{p^2}{m^2c^2} - ...\right) \\
&= mc^2 + \frac{p^2}{2m} - ... \,.
\end{aligned}
$$

If we use this hamiltonian on the Schrodinger equation,

$$
i\hbar \frac{\partial}{\partial t}\psi(\boldsymbol{x}, t) = +\sqrt{-\hbar^2c^2\nabla^2 + m^2c^4}\,\psi(\boldsymbol{x}, t),
$$

square both side's operators then we get the **Klein-Gordon** equation,

$$
-\hbar^2 \frac{\partial^2}{\partial t^2}\psi(\boldsymbol{x}, t) = (-\hbar^2c^2\nabla^2 + m^2c^4)\psi(\boldsymbol{x}, t).
$$

Without squaring both operators, there would be two problems:

1. Asymmetry of time and space. Time and space operators (in this case the differential) are separated, whereas in relativistic theory, time and space are one spacetime.
2. Non-locality in space. We already expanded hamiltonian using binomial expansion. We can see that the higher the order, there would be infinite number of differential of space (momentum operator).

To simplify this equation and bring the spacetime coordinate together, let us define a new differential operator.

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right) ,$$
$$\partial^\mu \equiv \frac{\partial}{\partial x_\mu} = \left(-\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right) .$$

The operator satisfies $\partial^\mu x^\nu = g^{\mu\nu}$ and relates with Lorentz transformation as $\bar{\partial}^\mu = \Lambda^\mu_\nu \partial^\nu$.

We can rewrite the Klein-Gordon as

$$\left(\partial^\mu \partial_\mu - m^2c^2/\hbar^2\right)\psi(x) = 0.$$

Suppose that we are looking at the equation in different inertial frame $\bar{x}$.

$$\begin{aligned}
\left(\bar{\partial}^\mu \bar{\partial}_\mu - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(g_{\mu\nu}\bar{\partial}^\mu \bar{\partial}^\nu - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(g_{\mu\nu}\Lambda^\mu_\rho \Lambda^\nu_\sigma \partial^\rho \partial^\sigma - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(g_{\rho\sigma}\partial^\rho \partial^\sigma - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 \\
\left(\partial^\mu \partial_\mu - m^2c^2/\hbar^2\right)\bar{\psi}(\bar{x}) &= 0 .
\end{aligned}$$

So, in both inertial frames, the wavefunctions satisfy the same Klein-Gordon equation.

Now that the Klein-Gordon equation satisfy conditions on the relativistic side, it actually violates the original Schrodinger equation. Notice how the time differential operator is in the first order but it is in the second order in Klein-Gordon equation. One of the prominent problem is that, the probability or norm of a state would then have some dependency on time (unconserved probability).

Dirac's attempt at solving this inconsistency starts with writing Schrodinger equation in the form

$$i\hbar\frac{\partial}{\partial t}\psi_a(x) = \left(-i\hbar c(\alpha^j)_{ab}\partial_j + mc^2\beta_{ab}\right)\psi_b(x)$$

where α^j and β are called the Dirac matrices in spin space and a and b are either 1 or 2. Dirac matrices are defined to obey these conditions

$$\{\alpha^j, \alpha^k\}_{ab} = 2\delta^{jk}\delta_{ab} , \quad \{\alpha^j, \beta\}_{ab} = 0 , \quad (\beta^2)_{ab} = \delta_{ab} .$$

Let us prove that these matrices must be even dimensional. First, we must find the eigenvalue of β using the third condition.

$$\begin{aligned}
\beta|b\rangle &= b|b\rangle \\
\beta^2|b\rangle &= b^2|b\rangle .
\end{aligned}$$

We know from the condition that β^2 in matrix form is just the identity matrix, so

$$b^2 = 1 \quad \rightarrow \quad b = \pm 1 .$$

Now all that is left is to make sure that $\text{Trace}(\beta) = 0$. To do that, let us consider $\{\alpha^j, \beta\}$.

$$\begin{aligned}
\{\alpha_j, \beta\} &= 0 \\
\alpha_j\beta + \beta\alpha_j &= 0 \\
\alpha_j\alpha_j\beta + \alpha_j\beta\alpha_j &= 0 \\
\text{Trace}\big(\alpha_j\alpha_j\beta\big) + \text{Trace}\big(\alpha_j\beta\alpha_j\big) &= 0 \\
\text{Trace}\big(\alpha_j^2\beta\big) &= 0 \ .
\end{aligned}$$

Now, consider the first condition. let $j = k$ and $a = b$ (diagonal matrix condition).

$$\begin{aligned}
2\alpha_j^2 &= 2 \\
\alpha_j^2 &= 1 \\
\alpha_j^2\beta &= \beta \\
\text{Trace}\big(\alpha_j^2\beta\big) &= \text{Trace}(\beta) = 0
\end{aligned}$$

For trace of α matrices, consider the second condition again. This time, instead of multiplying another alpha matrix, we multiply it with the inverse of β. The cyclical property of trace still applies. Then we can get that $\text{Trace}\big(\alpha_j\big) = 0$. Now, we have shown that the dirac matrices are even dimensional. As for the minimum dimension, Dirac matrices cannot be a 2×2 matrices. By working the anticommutator with the smallest matrices that satisfy $\{\alpha^j, \alpha^k\} = 2\delta^{jk}$, the Pauli matrices, we get that the only matrix capable of anticommuting with Pauli matrices is a zero matrix, or in another words, no "fourth" Pauli matrix exists. Therefore, their size must be 4×4.

From the traces of Dirac matrices, we can conclude that the trace of the hamiltonian ($H = c\alpha \cdot \boldsymbol{p} + mc^2\beta$) is also zero. So, the four eigenvalues are $+E(\boldsymbol{p})$, $+E(\boldsymbol{p})$, $-E(\boldsymbol{p})$, and $-E(\boldsymbol{p})$. Dirac's proposed solutions for the negative energy level: that by Pauli exclusion principles, those negative energy states are already occupied by electrons forming a sea of electrons.

To make a more generalized quantum theory in relativistic limits, we have to start over from the ground, but this time we put time and space in an equal footing. To accomplish that, one of the method is to just demote position (space) from operator to just parameter (like time in non-relativistic quantum mechanics).

Before getting into relativistic quantum mechancis we would like to "transform" ordinary quantum mechanics (non-relativistic) to field theory. We begin by introducing **quantum field** operator $a(\boldsymbol{x})$ and its conjugate $a^\dagger(\boldsymbol{x})$. For bosonic fields, the operators are subject to such rules:

$$[a(\boldsymbol{x}), a(\boldsymbol{x}')] = 0 \ , \quad \big[a^\dagger(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\big] = 0 \ , \quad \big[a(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\big] = \delta(\boldsymbol{x} - \boldsymbol{x}') \ .$$

Then, we can introduce the quantum field hamiltonian and state as

$$H = \int \mathrm{d}^3x a^\dagger(\boldsymbol{x})\left(-\frac{\hbar}{2m}\nabla^2+U(\boldsymbol{x})\right)a(\boldsymbol{x}) + \frac{1}{2}\int \mathrm{d}^3x\,\mathrm{d}^3y V(\boldsymbol{x}-\boldsymbol{y})a^\dagger(\boldsymbol{x})a^\dagger(\boldsymbol{y})a(\boldsymbol{y})a(\boldsymbol{x}) \ ,$$

$$|\psi,\, t\rangle = \int \mathrm{d}^3x_1 ... \,\mathrm{d}^3x_n \psi(\boldsymbol{x}_1, ..., \boldsymbol{x}_n; t)a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)|0\rangle \ .$$

We can apply the hamiltonian directly to the state ket. If we look at the form of hamiltonian, there are two elements of summation. We can call these as H_1 and H_2. We then can apply them separately to the state ket for the sake of simplicity. Before that, let us consider this equation.

$$\begin{aligned}
&a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = \big(a^\dagger(\boldsymbol{x}_1)a(\boldsymbol{x}) + \delta(\boldsymbol{x}-\boldsymbol{x}_1)\big)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = a^\dagger(\boldsymbol{x}_1)a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle + \delta(\boldsymbol{x}-\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad \vdots \\
&\quad = a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)a(\boldsymbol{x})|0\rangle + \sum_{j=1}^{n}\delta\big(x-x_j\big)\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = \sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_k)|0\rangle \ .
\end{aligned}$$

Now, we can consider the H_1 operator.

$$\begin{aligned}
&H_1|\psi,\, t\rangle \\
&\quad = \int \mathrm{d}^3x\,\mathrm{d}^3x_1 ... \,\mathrm{d}^3x_n a^\dagger(\boldsymbol{x})\left(-\frac{\hbar}{2m}\nabla^2+U(\boldsymbol{x})\right)\psi(\boldsymbol{x}_1, ..., \boldsymbol{x}_n; t)a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\quad = \int \mathrm{d}^3x\,\mathrm{d}^3x_1 ... \,\mathrm{d}^3x_n a^\dagger(\boldsymbol{x})K(\boldsymbol{x})\psi\left(\sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_k)|0\rangle\right) \\
&\quad = \sum_{j=1}^{n}\int \mathrm{d}x_1 ... \,\mathrm{d}x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\big(K_j\big(\boldsymbol{x}_j\big)\psi\big)|0\rangle \\
&\quad = \int \mathrm{d}x_1 ... \,\mathrm{d}x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right)\psi|0\rangle
\end{aligned}$$

Now, let us consider this equation first before moving on to H_2.

$$\begin{aligned}
&a(\boldsymbol{y})a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\qquad = \sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)a(\boldsymbol{y})\prod_{k\neq j}^{n}a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&\qquad = \sum_{j=1}^{n}\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\sum_{k\neq j}^{n}\delta(\boldsymbol{y}-\boldsymbol{x}_k)\prod_{l\neq k,j}^{n}a^\dagger(\boldsymbol{x}_l)|0\rangle
\end{aligned}$$

Let us consider H_2 operator.

$$
\begin{aligned}
&H_2|\psi,\, t\rangle \\
&= \int \mathrm{d}^3x\, \mathrm{d}^3y\, \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n V(\boldsymbol{x}-\boldsymbol{y})\psi a^\dagger(\boldsymbol{x})a^\dagger(\boldsymbol{y})a(\boldsymbol{y})a(\boldsymbol{x})a^\dagger(\boldsymbol{x}_1)a^\dagger(\boldsymbol{x}_2)...a^\dagger(\boldsymbol{x}_n)|0\rangle \\
&= \sum_{j=1}^{n}\sum_{k\neq j}^{n} \int \mathrm{d}^3x\, \mathrm{d}^3y\, \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n V(\boldsymbol{x}-\boldsymbol{y})\psi\delta\big(\boldsymbol{x}-\boldsymbol{x}_j\big)\delta(\boldsymbol{y}-\boldsymbol{x}_k)a^\dagger(\boldsymbol{x})a^\dagger(\boldsymbol{y}) \prod_{l\neq k,j}^{n} a^\dagger(\boldsymbol{x}_l)|0\rangle \\
&= \sum_{j=1}^{n}\sum_{k\neq j}^{n} \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\psi a^\dagger\big(\boldsymbol{x}_j\big)a^\dagger(\boldsymbol{x}_k) \prod_{l\neq k,j}^{n} a^\dagger(\boldsymbol{x}_l)|0\rangle \\
&= \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(\sum_{j=1}^{n}\sum_{k\neq j}^{n} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right)\psi|0\rangle \ .
\end{aligned}
$$

Putting all back together,

$$
\begin{aligned}
H|\psi,\, t\rangle &= H_1|\psi,\, t\rangle + \frac{1}{2}H_2|\psi,\, t\rangle \\
&= \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right) + \frac{1}{2}\sum_{j=1}^{n}\sum_{k\neq j}^{n} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right)\psi|0\rangle \\
&= \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right) + \sum_{j=1}^{n}\sum_{k=1}^{j-1} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right)\psi|0\rangle \ .
\end{aligned}
$$

Applying time operator to the state,

$$
i\hbar\frac{\partial}{\partial t}|\psi,\, t\rangle = \int \mathrm{d}^3x_1 ... \, \mathrm{d}^3x_n a^\dagger(\boldsymbol{x}_1)...a^\dagger(\boldsymbol{x}_n)\left(i\hbar\frac{\partial}{\partial t}\right)\psi|0\rangle \ .
$$

If we demand that the hamiltonial satisfy the Schrodinger equation, then

$$
\begin{aligned}
i\hbar\frac{\partial}{\partial t}|\psi,\, t\rangle &= H|\psi,\, t\rangle \\
i\hbar\frac{\partial\psi}{\partial t} &= \left[\sum_{j=1}^{n}\left(-\frac{\hbar}{2m}\nabla_j^2+U\big(\boldsymbol{x}_j\big)\right) + \sum_{j=1}^{n}\sum_{k=1}^{j-1} V\big(\boldsymbol{x}_j-\boldsymbol{x}_k\big)\right]\psi \ .
\end{aligned}
$$

Look at the integration from the state ket. Having wave function with space dependency and that creation operator commutes with each other, we can see that only symmetric ψ would survive the integration.

$$
\psi\big(...\boldsymbol{x}_i...\boldsymbol{x}_j...;t\big) = \psi\big(...\boldsymbol{x}_j...\boldsymbol{x}_i...;t\big) \ .
$$

Fermionic field, on the other hand, obeys these anticommutator relation:

$$
\{a(\boldsymbol{x}), a(\boldsymbol{x}')\} = 0 \ , \quad \big\{a^\dagger(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\big\} = 0 \ , \quad \big\{a(\boldsymbol{x}), a^\dagger(\boldsymbol{x}')\big\} = 0 \ .
$$

In contrast with the previous bosonic field, the fermionic wave equation's part that would survive the integration in state ket equation is the antisymmetric part.

$$
\psi\big(...\boldsymbol{x}_i...\boldsymbol{x}_j...;t\big) = -\psi\big(...\boldsymbol{x}_j...\boldsymbol{x}_i...;t\big) \ .
$$

There is another operator that counts the number of particles, that is

$$N = \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) a(\boldsymbol{x}) \ .$$

We can see how this commutes with previous hamiltonian for fields.

$$[N, H] = NH - HN \ ,$$

$$\begin{aligned}
NH &= \int \mathrm{d}^3 x' \, \mathrm{d}^3 x a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x' \, \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) \\
&= \int \mathrm{d}^3 x' \, \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') K(\boldsymbol{x}) a(\boldsymbol{x}') a(\boldsymbol{x}) + \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x' \, \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a(\boldsymbol{y}) a(\boldsymbol{x}) \\
&\quad + \int \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) \ , \\
HN &= \int \mathrm{d}^3 x \, \mathrm{d}^3 x' a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x \, \mathrm{d}^3 y \, \mathrm{d}^3 x' V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') \\
&= \int \mathrm{d}^3 x' \, \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{x}') K(\boldsymbol{x}) a(\boldsymbol{x}') a(\boldsymbol{x}) + \int \mathrm{d}^3 x a^\dagger(\boldsymbol{x}) K(\boldsymbol{x}) a(\boldsymbol{x}) \\
&\quad + \frac{1}{2} \int \mathrm{d}^3 x' \, \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a^\dagger(\boldsymbol{x}') a(\boldsymbol{x}') a(\boldsymbol{y}) a(\boldsymbol{x}) \\
&\quad + \int \mathrm{d}^3 x \, \mathrm{d}^3 y V(\boldsymbol{x} - \boldsymbol{y}) a^\dagger(\boldsymbol{x}) a^\dagger(\boldsymbol{y}) a(\boldsymbol{y}) a(\boldsymbol{x}) \ .
\end{aligned}$$

Thus,

$$[N, H] = 0 \ .$$